ANDERSON JUNIOR COLLEGE

2009 JC 1 H2 CHEMISTRY
PROMOTIONAL EXAMINATION

Name ……………………………………..…..(   ) **PDG ……….…..**

CHEMISTRY **9647**

Higher 2 **7 Oct 2009 (Wed)**

SECTION C: Multiple Choice **50 min**

Additional Materials: Data Booklet
Optical Answer Sheet (OAS)

READ THESE INSTRUCTIONS FIRST

1. Answer **ALL** questions on the OAS provided.

2. **Instructions for filling the OAS**:
   Write your name and PDG clearly.
   Write and shade all **7 digits** of your NRIC/FIN number.
   No reference letter is needed.

   All shading must be done with a 2B pencil.

IMPORTANT NOTE

The OAS will be collected at the end of 50 minutes.
Submit only the OAS. You may keep the question paper.

This document consists of **10** printed pages, including the cover page

Section C (Multiple Choice 30 marks)

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

1 The value of the Avogadro constant L is 6.02×10^{23} mol^{-1}. The mass, in g, of one molecule of propane (RMM = 44.0) is therefore

A $44.0 \times 6.02 \times 10^{23}$

B $\dfrac{6.02 \times 10^{23}}{44.0}$

C $\dfrac{44.0}{6.02 \times 10^{23}}$

D $\dfrac{1}{44.0 \times 6.02 \times 10^{23}}$

2 10 dm^3 of polluted air is passed through lime water so that all the carbon dioxide present is precipitated as calcium carbonate. The mass of calcium carbonate formed is 25 g. What is the percentage, by volume, of carbon dioxide in the air sample at s.t.p?

A 0.56 **B** 0.60 **C** 56 **D** 60

3 When potassium chlorate(V), $KClO_3$, is heated at its melting point, it disproportionates to potassium chlorate(VII), $KClO_4$, and potassium chloride. What is the maximum number of moles of potassium chlorate(VII) which could be produced from 0.1 mol of potassium chlorate(V)?

A 0.05 **B** 0.075 **C** 0.25 **D** 0.1

4 Which one of the following corresponds to the configuration of the three electrons of highest energy for the ground state of an element in Group III?

A $1s^2\ 2s^1$

B $2s^2\ 2p^1$

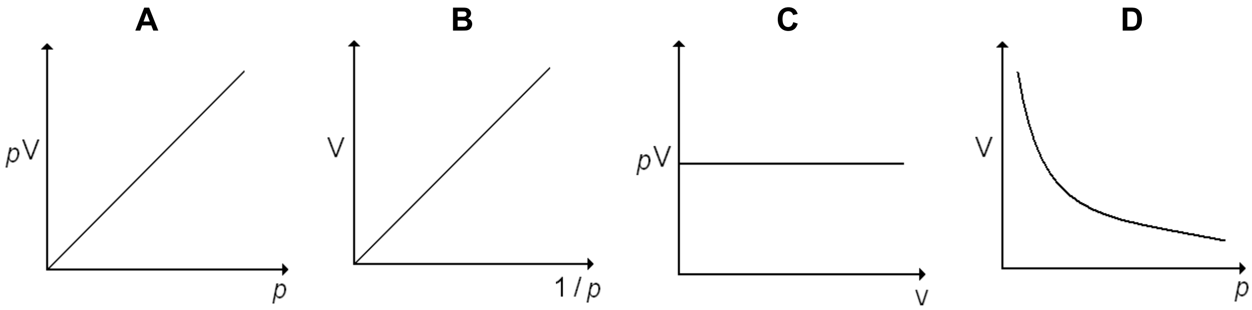
C $2s^1\ 2p^2$

D $3p^3$

5 A sample of gas in a flask contains 2 g of helium and 32 g of oxygen at 300 Pa. What is the partial pressure of helium in the flask?

A 60 Pa **B** 100 Pa **C** 150 Pa **D** 200 Pa

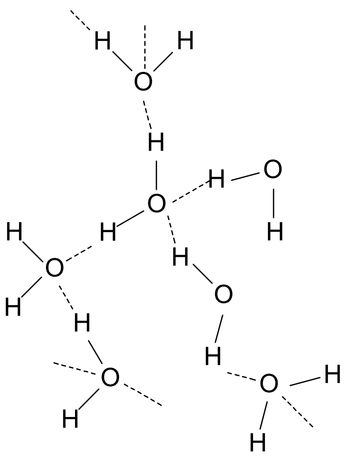
6 Which of the following graphs does **not** describe the relationship between volume and pressure for a fixed mass of an ideal gas under constant temperature?



7 Which of the following molecules or ions is **not** planar?

A $C_6H_5CH_3$ **B** BF_3 **C** IF_4^- **D** CO_3^{2-}

8 The diagram below shows the structure of part of a crystal of ice.



Which of the following statements is correct?

A All the bonds angles surrounding each oxygen atom are 120°.

B The hydrogen bonds are stronger than the O-H covalent bonds.

C The open structure of ice causes ice to be denser than water.

D Four electrons from each oxygen are involved in forming hydrogen bonds.

9 The data below refer to gas phase reactions at constant pressure.

$CH_3–CH_3 \rightarrow CH_3–CH_2\bullet + H\bullet \quad \Delta H^\ominus = +412\ kJ\ mol^{-1}$
$CH_3–CH_2\bullet \rightarrow CH_2=CH_2 + H\bullet \quad \Delta H^\ominus = +168\ kJ\ mol^{-1}$

From these data, calculate the enthalpy change for the following reaction.

$2CH_3–CH_2\bullet \rightarrow CH_3–CH_3 + CH_2=CH_2$

A + 244 kJ mol^{-1}

B – 122 kJ mol^{-1}

C – 244 kJ mol^{-1}

D – 580 kJ mol^{-1}

10 For which process is the enthalpy change **always** negative?

A burning an element in oxygen

B dissolving a compound in water

C forming an ion from an atom

D synthesising a compound from its elements

11 For which reaction is the change in entropy positive?
[Assume that all measurements are taken at 298 K and 1 atm.]

A $Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

B $H_2(g) + C_2H_4(g) \rightarrow C_2H_6(g)$

C $Mg(s) + \frac{1}{2} O_2(g) \rightarrow MgO(s)$

D $N_2O(g) \rightarrow N_2(g) + \frac{1}{2} O_2(g)$

12 A reaction is **always** spontaneous if its occurrence is

A exothermic with a decrease in entropy

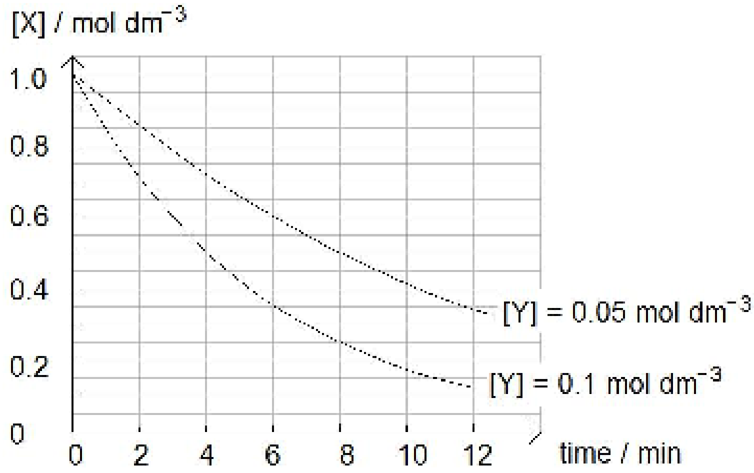
B exothermic with an increase in entropy

C endothermic with a decrease in entropy

D endothermic with an increase in entropy

13 An experiment is set up to measure the rate of the following reaction.

$$2\mathbf{X} + \mathbf{Y} \rightarrow \mathbf{W} + 2\mathbf{Z}$$



Which conclusion can be drawn from the above information?

A The reaction is zero order with respect to **X**.

B The rate of reaction is affected by the change in concentration of **X**.

C **Y** is not involved in the rate-determining step.

D The rate equation for the reaction can be written as: rate = $k\,[\mathbf{X}]^2[\mathbf{Y}]$.

14 The alkaline hydrolysis of RBr, where RBr = $(CH_3)_3CBr$, proceeds in two steps.

$$RBr \rightarrow R^+ + Br^- \quad \text{slow}$$
$$R^+ + OH^- \rightarrow ROH \quad \text{fast}$$

Which of the following rate equations is consistent with this scheme?

A rate = $k\,[RBr]\,[OH^-]$

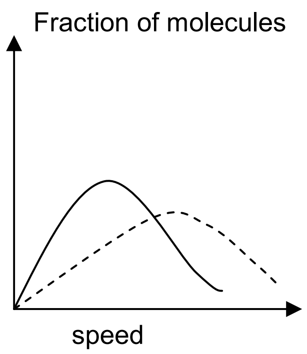
B rate = $k\,[R^+]\,[OH^-]$

C rate = $k\,[RBr]$

D It cannot be determined from the given information.

15 Which graph most accurately represents the distribution of molecular speeds in a gas at 300 K if the dotted curve represents the corresponding distribution for the same gas at 500K?

A



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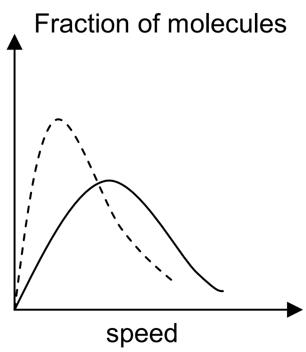

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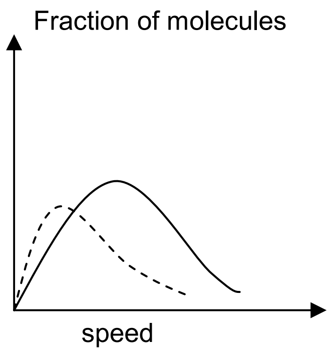

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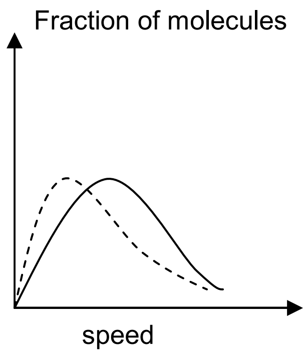
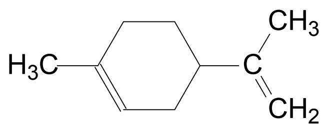


16 Limonene is an oil found in the peel of citrus fruits.



limonene

When limonene reacts with excess bromine at room temperature in the dark, how many **more** chiral centres than limonene does the product molecule possess?

A 2 **B** 3 **C** 4 **D** 5

17 How many alkenes (including both structural isomers and stereoisomers) are possible for C_5H_{10}?

A 5 **B** 6 **C** 7 **D** 10

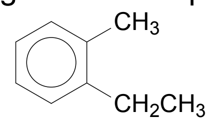
18 1,2–dibromo–3–chloropropane (DBCP) has been used in the control of earthworms in agricultural land. Which of the following would be the best synthesis of this compound?

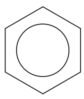
A $CH_2{=}CHCH_2Cl + Br_2 \rightarrow$ DBCP

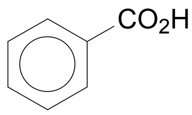
B $CH_3CHBrCH_2Br + Cl_2 \rightarrow$ DBCP + HCl

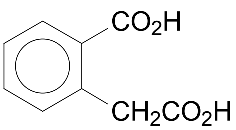
C $CH_2{=}CHCHBr_2 + HCl \rightarrow$ DBCP

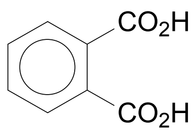
D $CH_3CH_2CH_2Cl + 2Br_2 \rightarrow$ DBCP + 2HBr

19 Which of the following formulae represents the organic compound formed when the aromatic compound  is heated under reflux with alkaline potassium manganate(VII) solution and the mixture is then acidified?

A 

B 

C 

D 

20 A mixture of 1 mol of chlorobenzene and 1 mol of 2-chloropropane is heated under reflux for several hours with 2 mol of potassium hydroxide in aqueous solution.

How many moles of chlorobenzene and 2-chloropropane are likely to remain?

	chlorobenzene	2-chloropropane
A	1	1
B	1	0
C	0	1
D	0	0

For each of the questions, **21** to **30**, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The correct responses **A** to **D** should be selected on the basis of

A	B	C	D
1,**2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

21 Which statements about the complete combustion of an alkene, C_nH_{2n}, in oxygen are correct?

1 The volume of oxygen required is directly proportional to the number of carbon atoms present in the molecule.

2 The volume of gas produced at 25 ^{o}C is the same as for the complete combustion of an alkane with the same number of carbon atoms per molecule.

3 At 120^{o}C, the volume of steam produced is always twice the volume of carbon dioxide.

22 Which of the following statements about the reaction

$$IO_3^- (aq) + 2I^- (aq) + 6H^+ (aq) + 6\ Cl^- (aq) \rightarrow 3ICl_2^- (aq) + 3H_2O\ (l)$$

are correct?

1 Disproportionation occurs in the reaction.

2 The oxidation number of iodine in the iodide ion, I^- (aq), changes from -1 to +1.

3 The oxidation number of iodine in the iodate ion, IO_3^-(aq), changes from + 5 to +1.

23 The first ionisation energy of Period 3 elements generally increases across the period. Which factors explain why the value of the first ionisation energy of sulfur is lower than that of phosphorus?

1 repulsion between the pair of 3p electrons

2 greater shielding by inner electrons

3 increase of principal quantum shell

The correct responses **A** to **D** should be selected on the basis of

A	B	C	D
1,**2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

24 Which of the following molecules are polar?

1 $BeCl_2$

2 CH_2F_2

3 SF_4

25 Which of the following statements about the molecule $HC{\equiv}CCH_3$ are correct?

1 All the carbon atoms lie on the same plane.

2 There are a total of 3σ and 2π bonds in the molecule.

3 The molecule dissolves readily in water.

26 Which reactions represent standard enthalpy changes at 298 K?

1 $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

2 $6C(g) + 6H(g) \rightarrow C_6H_6(g)$

3 $CH_3OH(g) + 3/2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

27 Which of the following may affect the rate constant for a reaction?

1 change in temperature

2 change in concentration

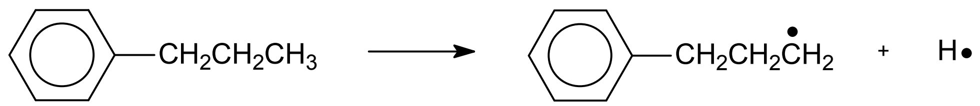
3 change in pressure

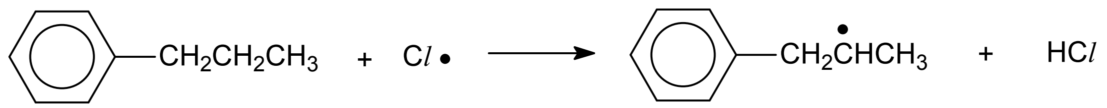
The correct responses **A** to **D** should be selected on the basis of

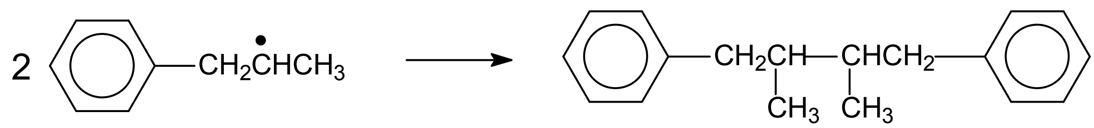
A	B	C	D
1,**2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

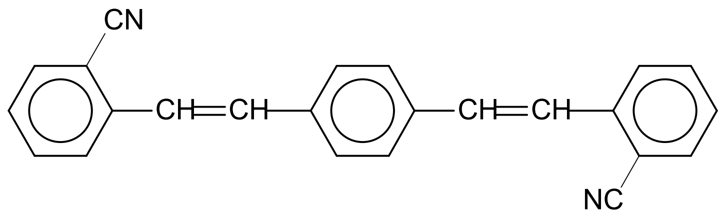
28 Which of the following reactions occur when propylbenzene is mixed with Cl_2 in the presence of sunlight?

1 

2 

3 

29 A compound used as an optical brightener in detergent has the following formula.



Which of the following statements are correct?

1 It can be oxidised by cold alkaline potassium manganate(VII).

2 It can undergo both electrophilic addition and electrophilic substitution with Br_2.

3 It exhibits geometric isomerism.

30 Which compounds may be prepared from $C_6H_5CHBrCH_3$ by the action of sodium hydroxide under different conditions?

1 $C_6H_5CO_2Na$

2 $C_6H_5CH(OH)CH_3$

3 $C_6H_5CH{=}CH_2$